

Python for Researchers — Day 3

Pandas Cheatsheet & Concept Guide

Quick Reference — All Methods Covered Today

Method	What it does
<code>pd.read_csv()</code>	Load a CSV file into a DataFrame
<code>df.info()</code>	Summary: shape, dtypes, null counts
<code>df.head()</code> / <code>df.tail()</code>	Preview top or bottom rows
<code>df.dtypes</code>	Data type of each column
<code>df.copy()</code>	Make a safe copy before editing
<code>df.drop(columns=[...])</code>	Remove columns by name
<code>df.drop(index)</code>	Remove a row by index label
<code>df["col"]</code>	Isolate one column (returns Series)
<code>df[["col1", "col2"]]</code>	Isolate multiple columns (returns DataFrame)
<code>df.loc[condition]</code>	Filter rows by value or condition
<code>df.iloc[rows, cols]</code>	Filter by integer position
<code>df.isin([values])</code>	Check if values are in a list
<code>df.value_counts()</code>	Count occurrences of each value
<code>df.reset_index()</code>	Convert Series index back to columns
<code>df.sort_values(col)</code>	Sort DataFrame by a column
<code>df.groupby(col)[...].mean()</code>	Group rows and aggregate data
<code>df.astype({col: type})</code>	Convert a column's data type
<code>pd.to_numeric(...)</code>	Convert to number;
<code>series.notna()</code>	True where values are NOT NaN
<code>series.to_list()</code>	Convert Series to a Python list